

SAVANNAH CHAPPUS

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PROFILE

Animator, designer, mathematician, and developer with expertise in 3D animation, visualization, simulation, UI/UX, and logo design. STEM background provides a unique ability to merge technical rigor with creative storytelling, producing compelling, data-driven visual experiences.

Proficient in Adobe Creative Suite, (Illustrator, InDesign, After Effects), Blender, and scripting in Python and HTML. Self-motivated and resourceful, with a proven ability to take initiative and learn quickly. Eager to expand professional knowledge and deliver innovative, high-quality results through technical expertise, problem-solving, and creative vision.

EDUCATION

Master of Science: Computer Science
May 2025 — Northern Arizona University

Bachelor of Fine Arts: Visual Communication (Motion Design emphasis)
May 2023 — Northern Arizona University

Bachelor of Science in Mathematics
May 2023 — Northern Arizona University

KEY QUALIFICATIONS

Animation, Simulation & Visualization

- Designed and developed the Draugr FUI system animation — a planetary system built based on real data, utilizing Blender's node editor for texturing, and After Effects for UI and animated overlays.
- Produced a promotional animation for a local company, including character design, 3D modeling, rigging, animation, and scene composition.
- Built interactive augmented reality prototypes in Unity, blending design and development for immersive user experiences.
- Designed and developed basic computer games in Unity, including an apple picker, space shooter, and tower defense game, combining graphics programming, physics, and user interaction design.

Research & Technical Development

- Designed and developed the ADAPT pipeline, leveraging machine learning to analyze over 32,000 asteroid observations, detect anomalies in asteroid behavior, and visualize large-scale datasets [github.com/niteflyunicorns/ADAPT].
- Contributed to the Solar System Notification Alert Processing System (SNAPS), developing algorithms for detecting asteroid behavior changes.
- Conducted an exploratory research analysis on the Yamma board game utilizing computing and data visualization concepts to support an in-depth combinatorial game theory analysis, including visualizing complex graph networks using Python.
- Simulated a comprehensive offensive security analysis and report on a real organization, employing ethical hacking techniques to research potential security vulnerabilities and propose actionable solutions.
- Applied high-performance computing (HPC) techniques to research in deep learning and astroinformatics, leveraging available resources for large-scale data analysis.
- Conducted research on GPU optimizations for DBSCAN, focusing on performance optimization and advanced data visualization for clustering results.
- Advanced the SOILWAT2 ecosystem simulation model by programming new features in C and R, debugging legacy code, and managing contributions through Git/GitHub [github.com/DrylandEcology/SOILWAT2].

Design & Creative Work

- Designed an app prototype for the GRBY SQRD logic game, and initiated development in Android Studio.
- Created promotional materials — including flyers and stickers — for a local roller derby league as a volunteer designer.

Mentorship & Team Management

- Provided group tutoring and resource development for over 500 students in advanced mathematics, supporting and encouraging good study skill development and exam prep.
- Mentored undergraduate capstone teams, coordinating between clients, faculty, and students, delegating tasks, and overseeing project timelines.
- Co-led the SNAPS astroinformatics bootcamp, mentoring 15+ students in sprint-style development projects.
- Managed a small team of undergraduate graders, ensuring consistent grading quality, detailed feedback, and efficient workflow.

PROFESSIONAL EXPERIENCE

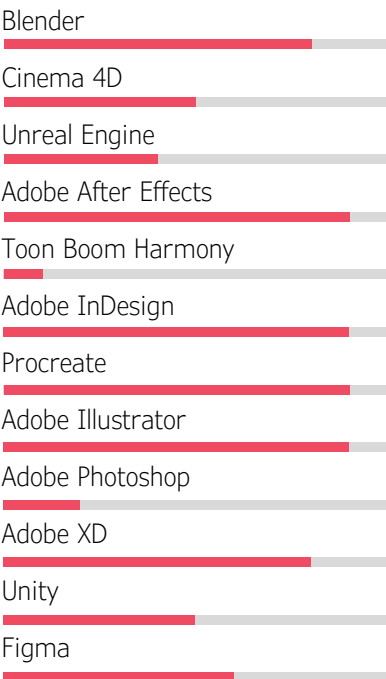
- 3D Animator:** Freelance | Aug 2023 – Present
- Student Researcher:** Northern Arizona University | Jan 2023 – May 2023, Apr 2024 – Present
- Computer Science Capstone Mentor:** Northern Arizona University | Jan 2025 – May 2025
- Graduate Teaching Assistant:** Northern Arizona University | Jan 2025 – May 2025
- Ecological Programmer:** United States Geological Survey | Oct 2023 – May 2024
- Peer Tutor:** Northern Arizona University Academic Success Centers | Aug 2021 – May 2023

SELECTED PROJECTS

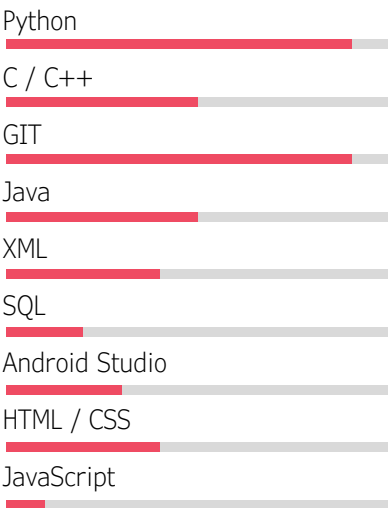
- Personal Portfolio | Web Design & Development**
Designed and built a personal portfolio website, entirely self-taught using HTML, CSS, and JavaScript to showcase design work.
- SNAPS Website | Web Design**
Designed the new SNAPS project website, implementing a user-focused dashboard, and streamlined data access, with a modern, intuitive interface.
- Mealie | Open Source Project**
Contributed to an open-source, self-hosted recipe management platform (Vue.js, Python), improving user experience by adding contextual messaging to reduce confusion and enhance interface clarity. [github.com/mealie-recipes/mealie].
- KHES & SGAFT Logos | Logo Design**
Updated Peter Konjoian’s existing logo to meet new business decisions and created a new logo for an upcoming project.
- Foodicons | Icon Design**
Designed several food/agriculture based icons which were selected as part of a crowd sourced icon design project organized by The Noun Library and The Lexicon.
- TXG Logo | Logo Design**
Logo redesign and update for the Tribal Exchange Network Group (TXG).

SKILLS

ANIMATION & DESIGN



COMPUTING



REFERENCES

- Dana Ernst**
Professor | dana.ernst@nau.edu
- Mike Gowanlock**
Advisor | michael.gowanlock@nau.edu
- David Trilling**
Advisor | david.trilling@nau.edu